CORE PURE / POLAR COORDINATES

Tangents to Polar Curves

Questions by topic 8 questions · 94 marks

Questions

Question 1	*6 marks*	2
Question 2	*6 marks*	3
Question 3	*17 marks*	4
Question 4	*17 marks*	5
Question 5	*10 marks*	6
Question 6	*10 marks*	7
Question 7	*11 marks*	8
Question 8	*17 marks*	9

1. The curve C has polar equation

$$r = 3\sin 2\theta, \qquad 0 \leq \theta \leq \frac{\pi}{2}$$

At the non-pole point P on C, the tangent to C is perpendicular to the initial line.

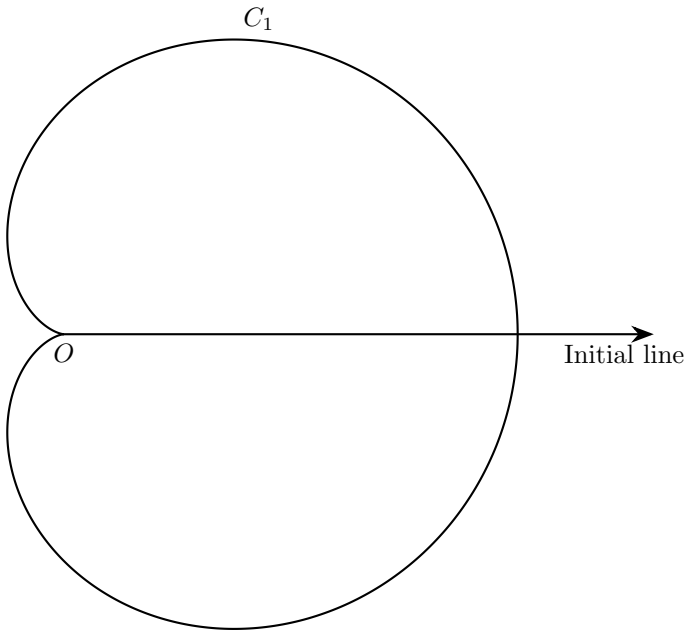
Given that O is the pole, find the exact length of the line OP. **[6]**

2. The curve C has polar equation

$$r = 2\sin 3\theta, \qquad 0 \leq \theta \leq \frac{\pi}{3}$$

At the non-pole point P on C, the tangent to C is parallel to the initial line.

Given that O is the pole, find the exact length of the line OP. **[6]**



3. The diagram shows the sketch of a curve C_1.

The polar equation of the curve C_1 is

$$r = 2(1 + \cos\theta), \qquad -\pi \leq \theta \leq \pi$$

(a) Find the area of the region bounded by the curve C_1. **[4]**

(b) The curve C_2 whose polar equation is

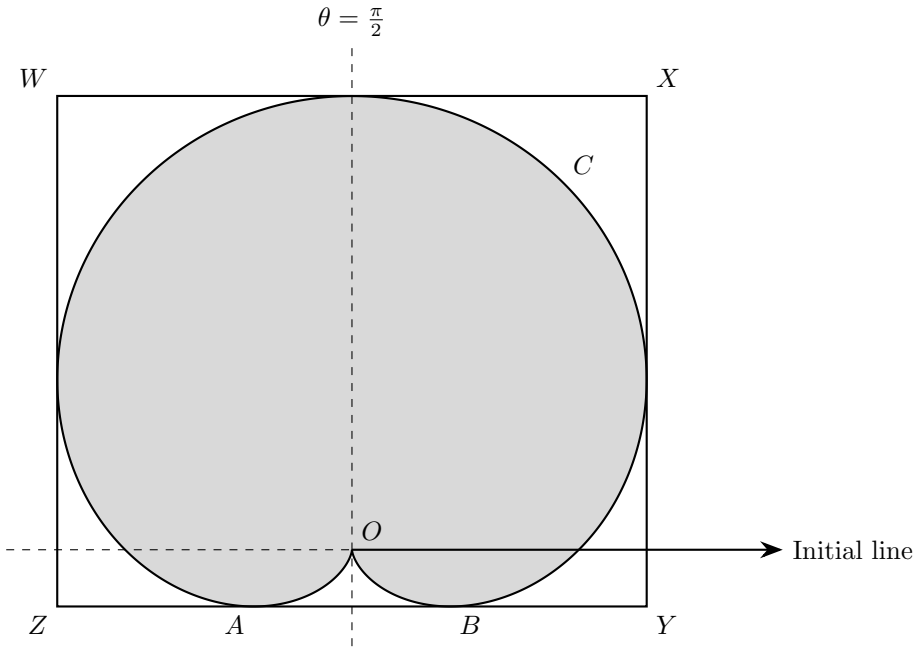
$$r = \sin\theta, \qquad 0 \leq \theta \leq \pi$$

intersects the curve C_1 at the pole O and at the point A. The straight line drawn through A perpendicular to the initial line intersects the part of C_1 above the initial line again at the point B.

(i) Find the polar coordinates of A. **[4]**

(ii) Show that the length of OB is $\dfrac{6}{5}$. **[6]**

(iii) Find the length of AB, giving your answer to three significant figures. **[3]**



4. The figure shows a sketch of the cardioid C with equation

$$r = a(1 + \sin\theta), \qquad 0 \leq \theta < 2\pi$$

Also shown are the tangents to C that are parallel and perpendicular to the initial line. These tangents form a rectangle $WXYZ$.

(a) Find the area of the finite region, shaded in the figure, enclosed by C. **[4]**

(b) Find the values of θ for which the tangent to C is parallel to the initial line. Hence find the polar coordinates of the points A and B, where ZY touches C. **[6]**

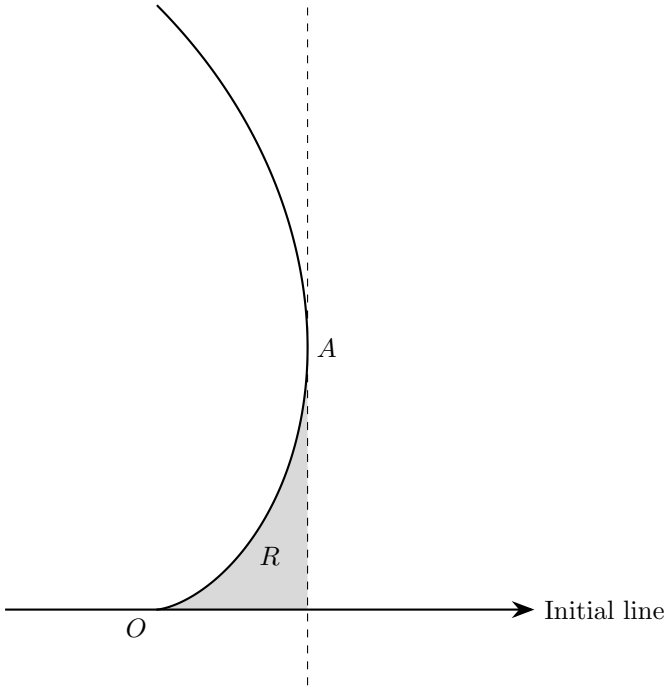
(c) Hence show that the length of WZ is $\frac{9a}{4}$. **[2]**

Given that the length of WX is $\frac{3\sqrt{3}a}{2}$

(d) find the area of the rectangle $WXYZ$. **[2]**

A logo is modelled by the cardioid C, where $a = 8\,\text{cm}$. The logo is cut from the rectangular card $WXYZ$, shown in the figure.

(e) Find a numerical value for the area of card wasted in making the logo. **[3]**



5. The diagram above shows a sketch of the curve C with equation

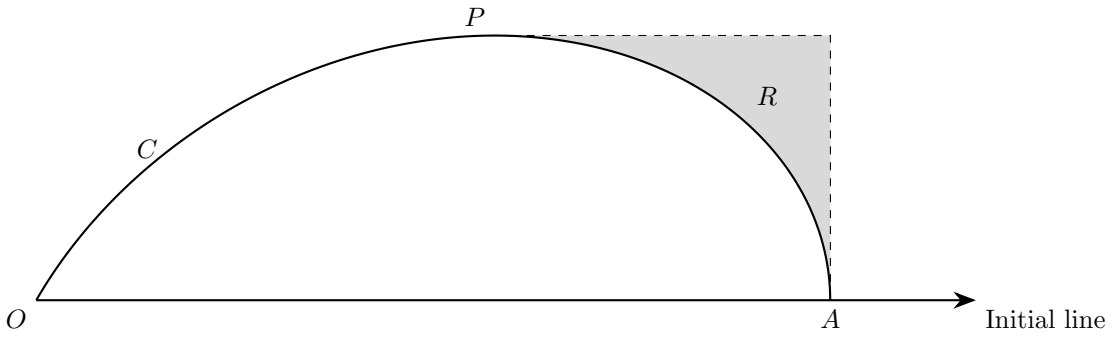
$$r = 4(1 - \cos\theta) \qquad 0 \leq \theta \leq \frac{\pi}{2}$$

The diagram above also shows the tangent to C at the point A, where A is not the pole.
This tangent is perpendicular to the initial line.

(a) Use differentiation to prove that the polar coordinates of A are $\left(2, \frac{\pi}{3}\right)$. **[4]**

The finite region R, shown shaded in the diagram above, is bounded by C, the tangent at A and the initial line.

(b) Use calculus to show that the exact area of R is $\frac{1}{2}(15\sqrt{3} - 8\pi)$. **[6]**



6. The curve C shown in the figure above has polar equation

$$r = 1 + 2\cos 2\theta, \qquad 0 \leq \theta \leq \frac{\pi}{3}$$

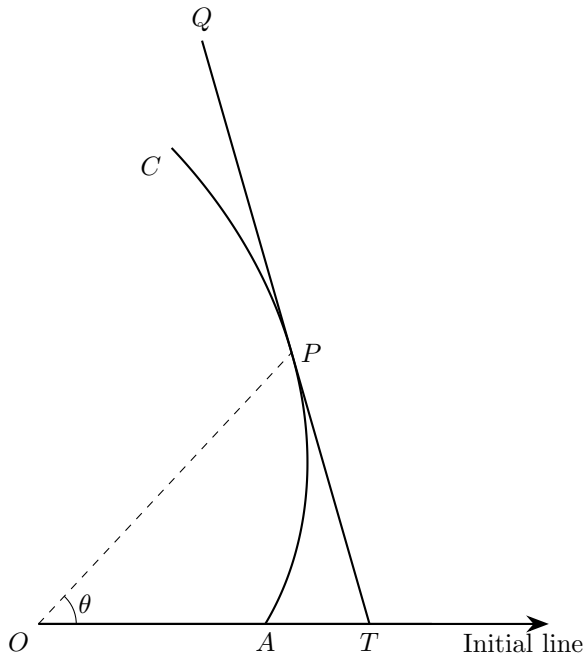
At the point P on C, the tangent to C is parallel to the initial line.

(a) Show that P has polar coordinates $\left(2, \frac{\pi}{6}\right)$. **[4]**

The point A on C has polar coordinates $(3, 0)$.

The finite region R, shown shaded in the figure above, is bounded by the arc AP of C, the tangent to C at P and the line through A parallel to $\theta = \frac{\pi}{2}$.

(b) Find the exact area of R. **[6]**



7. The diagram shows part of a spiral C.

The point P has polar coordinates (r, θ), where $0 \leq \theta \leq \frac{\pi}{2}$.

The tangent at P is the line TPQ. The point A lies on C and on the initial line, with $OA = 2$.

Let ψ be the angle that TPQ makes with the initial line.

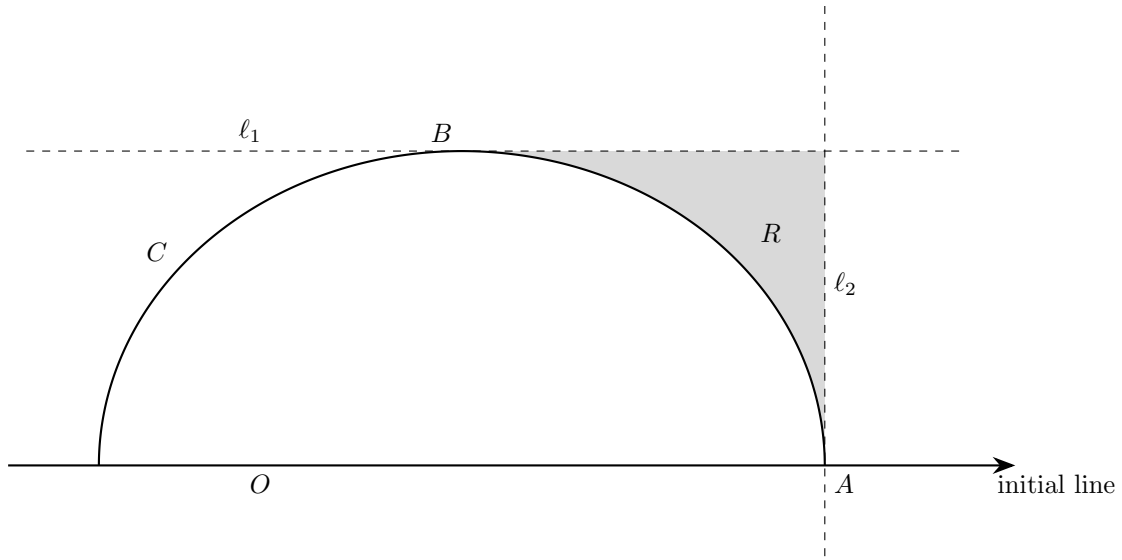
(a) By expressing the Cartesian coordinates of P in terms of r and θ, show that

$$\tan\psi = \frac{\frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta}{\frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta} \qquad \textbf{[4]}$$

(b) The spiral is equiangular: at every point P, the acute angle between OP and TPQ is a constant α, where $0 < \alpha < \frac{\pi}{2}$.

Use the result of part (a) to show that the polar equation of C is

$$r = 2e^{(\cot\alpha)\theta} \qquad \textbf{[7]}$$



8. The figure above shows a sketch of the curve C with equation

$$r = \frac{6}{2 - \cos\theta} \qquad 0 \leq \theta \leq \pi$$

Given that C meets the initial line at the point A, as shown in the figure above

(a) write down the polar coordinates of A. **[1]**

The line ℓ_1, also shown in the figure above, is the tangent to C at the point B and is parallel to the initial line.

(b) Use calculus to determine the polar coordinates of B. **[6]**

The line ℓ_2, also shown in the figure above, is the tangent to C at A and is perpendicular to the initial line.

The region R, shown shaded in the figure above, is bounded by C, ℓ_1 and ℓ_2.

(c) By first obtaining a Cartesian equation for C, use algebraic integration to find the exact area of R, giving your answer in the form $\sqrt{3}(p + q\pi)$ where p and q are constants to be determined. **[10]**